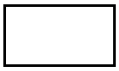
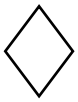
12. Entity Relationship Diagram

This section covers the theoretical description of the Entity-Relationship design of the Medical Appointment & Prescription System, including a description of the notations used, database entities, attributes, and cardinalities.

12.1 ER Components & Symbols

Symbol / Element Name	Visual Representation	Description
Entity		Represents a core database table or real-world object within the system(e.g., USERS, PATIENTS, DOCTORS).
Relationship		Represents the operational association or interaction linking two entities together(e.g., Has Profile, Books, Attends).
Attribute		Represents the specific properties, data fields, or columns belonging to an entity(e.g., username, age, specialization).
Primary key		Identifies the unique attribute used to uniquely distinguish each record within an entity table(e.g., user_id, patient_id).
One to one	1:1	Defines a strict mapping constraint where a single record in one entity relates to exactly one record in another(e.g., Users to Patients).
One to many	1:N	Defines a mapping constraint where a single record in an entity relates to multiple records in another(e.g., Patients to Appointments).

12.2 Entity and Attributes

i. Users

- **Purpose:** Stores the data related to central authentication such as usernames, the secure hashed passwords, and roles of all the users of the system.
- **Attributes**
  - user_id (Primary Key, INT)
  - username (VARCHAR)
  - password_hash (VARCHAR)
  - role (VARCHAR)

ii. Patients

- **Purpose:** Stores the contact and personal information of all the people eligible to receive services through the system.
- **Attributes:**

  - patient_id (Primary Key, INT)
  - full_name (VARCHAR)
  - age (INT)
  - gender (VARCHAR)
  - contact_number (VARCHAR)
  - residential_address (TEXT)
  - blood_group (VARCHAR)
  - registration_date (DATETIME)

iii. Doctors

- **Purpose:** Stores the personal and professional information of all the medical personnel such as their specializations, contact information, and schedule.
- **Attributes:**

  - doctor_id (Primary Key, INT)
  - full_name (VARCHAR)
  - specialization (VARCHAR)
  - contact_number (VARCHAR)
  - working_hours (VARCHAR)
  - consultation_fee(DECIMAL)
  - availability_status (BOOLEAN)

iv. Appointments

- **Purpose:** Stores the schedule of doctors' appointments including the date and time of each appointment
- **Attributes:**

  - appointment_id (Primary Key, INT)
  - appointment_date (DATE)
  - appointment_time (TIME)
  - status (VARCHAR)
  - created_at (DATETIME)

v. Prescriptions

- **Purpose**: Stores the medical treatment records of all the patients in electronic form including the prescribed drugs and notes made by doctors during each visit.
- **Attributes:**

  - prescription_id (Primary Key, INT)
  - medicines_and_dosage (TEXT)
  - clinical_notes (TEXT)
  - prescription_date (DATETIME)

vi. Medical_histories

- **Purpose:** Captures detailed clinical notes, symptoms, diagnoses, and treatment progress during physicians' consultations with patients
- **Attributes:**

- history_id (Primary Key, INT)
- recorded_at (DATETIME)
- symptoms (TEXT)
- diagnosis (TEXT)
- treatment_notes (TEXT)

vii. Medicine_reminders

- **Purpose**: Manages dosage alarms and tracking of medication to ensure patients that they take the dose of medication according to the prescribed schedule.
- **Attributes :**

  - reminder_id (Primary Key, INT)
  - medicine_name (VARCHAR)
  - dosage (VARCHAR)
  - scheduled_at (DATETIM|E)
  - snoozed_until (DATETIME)
  - status (VARCHAR)
  - interval_hours(INT)

viii. Medicine_dose_histories

- **Purpose**: Tracks patient medication adherence by recording and logging if scheduled medicine doses are given or skipped.
- **Attributes :**

  - history_id (Primary Key, INT)
  - medicine_name (VARCHAR)
  - dosage (VARCHAR)
  - scheduled_at (DATETIM|E)
  - given_at(DATETIME)
  - status (VARCHAR)

12.3 Relationship

Entity one	Relation	Entity two	Relation Linkage
Users	Register	Patients	The Patients table has a foreign key (registered_by) that points to the primary key (user_id) of the Users table, indicating who among the staff or admin created a patient record..
Users	has profile	Doctors	The Doctors table contains a foreign key (user_id) linked to the primary key (user_id) in the Users table.
Patients	Books	Appointments	The Appointments table contains a foreign key (patient_id) linked to the primary key (patient_id) in the Patients table
Doctors	Attends	Appointments	.The Appointments table contains a foreign key (doctor_id) linked

			to the primary key (doctor_id) in the Doctors table.
Appointments	Generates	Prescriptions	The Prescriptions table contains a foreign key (Appointment_id) linked to the primary key (appointment_id) in the Appointments
Patients	Receives	Prescriptions	The Prescriptions table contains a foreign key (patient_id) linked to the primary key (patient_id) in the Patients table.
Doctors	Issues	Prescriptions	The Prescriptions table contains a foreign key (doctor_id) linked to the primary key (doctor_id) in the Doctors table.
Patients	Has History	Medical_histories	The medical_histories table contains a foreign key (patient_id) that links to the patients.patient_id, establishing a relationship where a patient accumulates multiple historical clinical records.
Doctors	Records	Medical_histories	The medical_histories table contains a foreign key (doctor_id) that links to the doctors.doctor_id. This denotes which medical practitioner diagnosed and authored the clinical notes.
Patients	Receieves	Medicine_reminders	The medicine_reminders table contains a foreign key (patient_id) that links to the patients.patient_id. This maintains a relationship for tracking personal medication alerts.
Prescriptions	Triggers	Medicine_reminders	The medicine_reminders table contains a foreign key (prescription_id) that links to the prescriptions.prescription_id. This establishes a relationship where a single prescription generates specific scheduled dosage alerts.
Patients	Has History	Medicine_dose_histories	The medicine_dose_histories table has a foreign key (patient_id) that references patients.patient_id, establishing a relationship with the patient's adherence log.
Medicine_reminders	Has Records	Medicine_dose_histories	The medicine_dose_histories table has a foreign key (reminder_id) that references medicine_reminders.reminder_id, connecting the dose tracking

			record to the corresponding medicine reminder.

12.4 Cardinality Mapping

Entity One	Cardinality	Entity Two	Mechanism
Users	1:N	Patients	A single staff or administrative user can register multiple patient profiles over time (registered_by foreign key linkage)
Users	1:1	Doctors	Each professional doctor's data profile is tied to a singular administrative user's account login.
Patients	1:N	Appointments	Every single registered patient is allowed and able to make multiple appointment slots for consultations.
Doctors	1:N	Appointments	A single medical doctor arranges and schedules multiple appointment slots for consultations within their working hours.
Appointments	1:1	Prescriptions	Every finished appointment slot for consultation generates a singular digital prescription record.
Patients	1:N	Prescriptions	A single patient holds multiple historical prescription records, relating to their medical history.
Doctors	1:N	Prescriptions	A medical doctor issues multiple prescription records to multiple patients, within their working hours.
Patients	1:N	Medical_histories	A single patient record contains many historical clinical records over several patient visits connected by patient_id
Doctors	1:N	Medical_histories	An attending doctor creates and maintains many medical history records for different

			patients linked by doctor_id.
Patients	1:N	Medicine_reminders	A single patient record deals with many dosage alerts for ongoing treatments connected through patient_id.
Prescriptions	1:N	Medicine_reminders	One individual prescription slip creates many scheduled medication reminders connected using prescription_id.
Patients	1:N	Medicine_dose_histories	One medicine reminder is associated to multiple historical dose execution logs as time passes individual doses are taken, skipped, or monitored
Medicine_reminders	1”N	Medicine_dose_histories	A single patient profile consists of multiple medication adherence and dose history logs for various prescriptions and reminders.

12.5 Diagram

